

Lab Report of Group 1 (Wörner, Velez, Northe)

The source code and the binaries from the spectrum analyzer from this report can be found in our Github repository.

Our used imports

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
```

Our used helper functions to keep the code short

4.1 Task 1: Amplitude Modulation

Used Settings on Signal Analyzer

$$VBW = 100Hz \text{ } RBW = 3kHz \text{ } f_c = 2GHz \text{ } N_{sample} = 400$$

The following images show the configuration used to capture and visualize the AM signal:

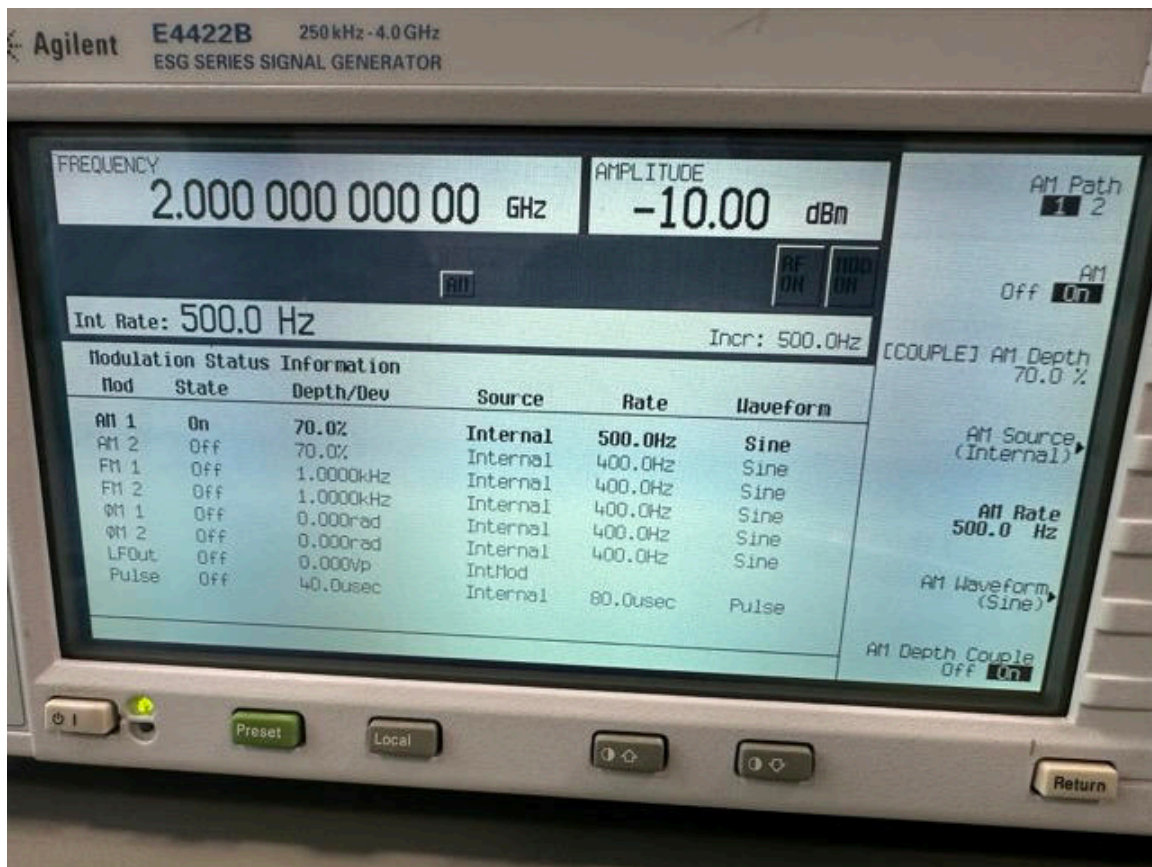


Figure 1: Signal generator configured for a 2 GHz carrier frequency, 500 Hz modulation frequency, and 70% AM depth.

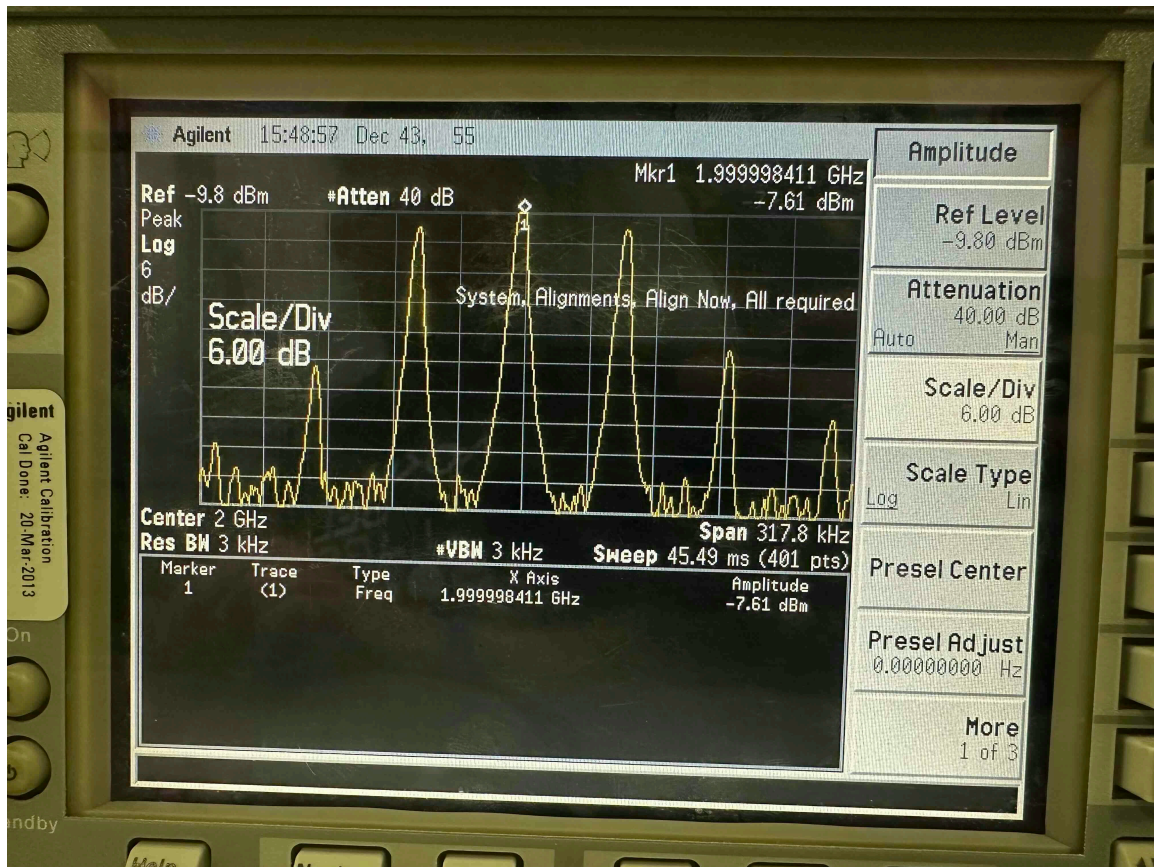


Figure 2: Spectrum analyzer displaying the AM signal with visible carrier and sidebands. The sideband spacing is consistent with the 500 Hz modulation frequency.

Observation: We noticed more than one sideband pair, when setting the output power of the signal generator to -10 dBm . This was not present when lowering it to -50 dBm as one can see in the screenshots below. We suspect that there are non-linear behaviors in either the signal generators, as we tried with both available to exclude any faults in one of them, or the signal analyzer. In the screenshot with the lower signal level, we can observe the expected spectrum with one strong main carrier peak and two symmetric and equally powered sideband peaks.

Observation: To investigate the effect of signal amplitude on spectral visibility, the output power of the signal generator was reduced from -10 dBm to -50 dBm . All other settings (carrier frequency, modulation frequency, and modulation index) were kept the same.

At this much lower power level, the sidebands became significantly weaker, and only two narrow spikes remain in the spectrum. These correspond to the AM modulation sidebands.

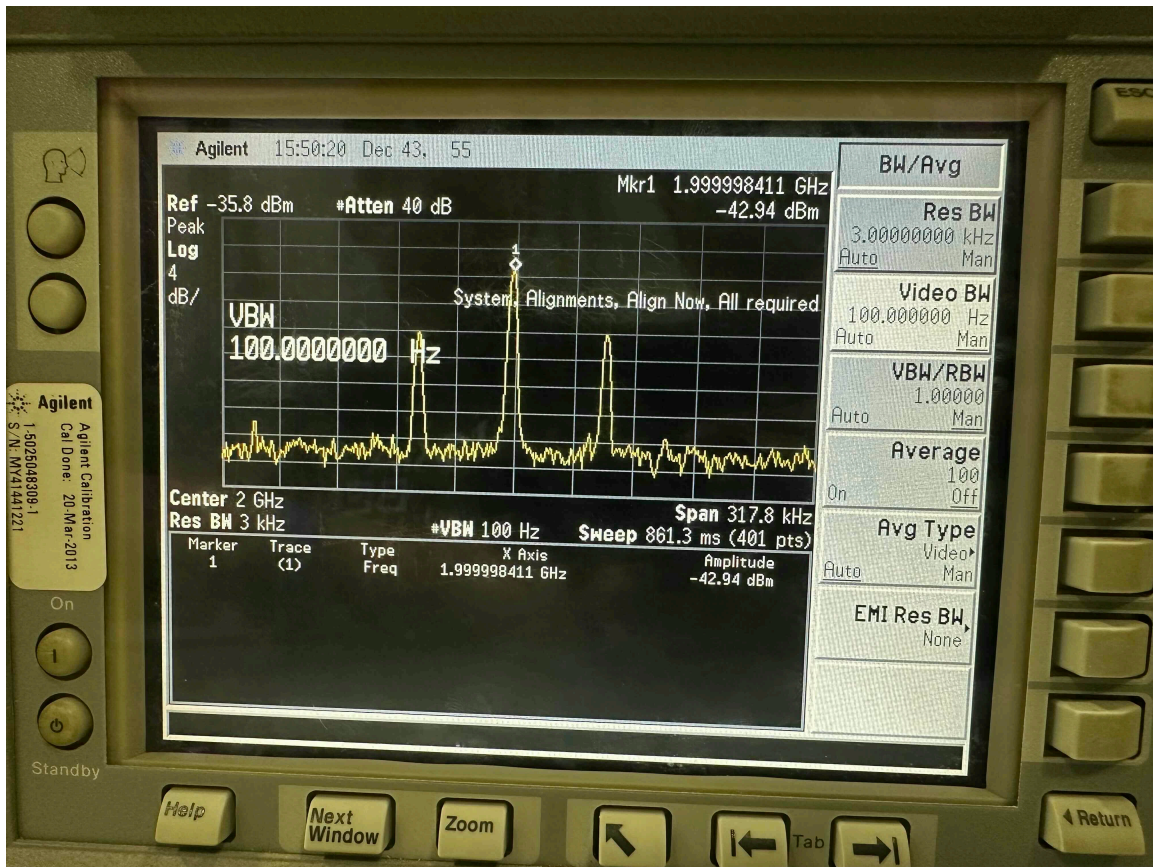


Figure 3: Spectrum analyzer display at -50 dBm input power. Only the AM modulation sidebands are visible.

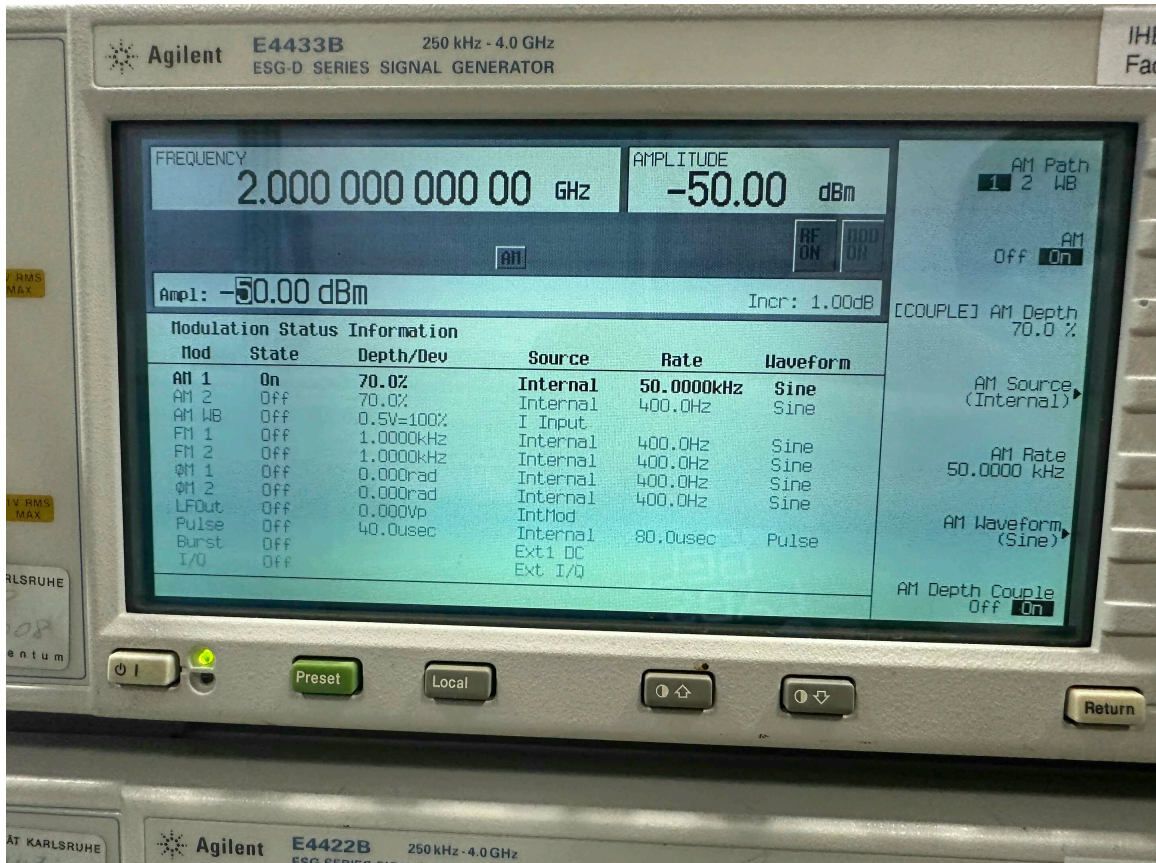


Figure 4: Signal generator configured for 2 GHz carrier, 500 Hz modulation, 70% depth, and output power set to -50 dBm.

4.1.3 Complete the following table by writing down the sideband attenuation and sideband frequencies for different values of the modulation index:

```
# Define modulation indices
modulation_indices = [0.001, 0.01, 0.25, 0.7, 1.0]

# Compute theoretical sideband levels in dB
def sideband_level(mu):
    return 20 * np.log10(mu / 2)

# Create dataframe
data = {
    "Modulation index ( $\mu$ )": modulation_indices,
    "Theoretical sideband level (dB)": [round(sideband_level(mu), 2) for mu in
modulation_indices]
}
```

```
df = pd.DataFrame(data)

# Display the table
print(df)
```

	Modulation index (μ)	Theoretical sideband level (dB)
0	0.001	-66.02
1	0.010	-46.02
2	0.250	-18.06
3	0.700	-9.12
4	1.000	-6.02

Modulation index (μ)	Sideband spacing	Measured sideband level (dB)	Theoretical sideband level (dB)
0.001	50.051 kHz	-61.63	-66.02
0.01	50.051 kHz	-43.40	-46.02
0.25	50.051 kHz	-16.41	-18.06
0.7	50.051 kHz	-7.44	-9.03
1.0	50.051 kHz	-4.90	-6.02

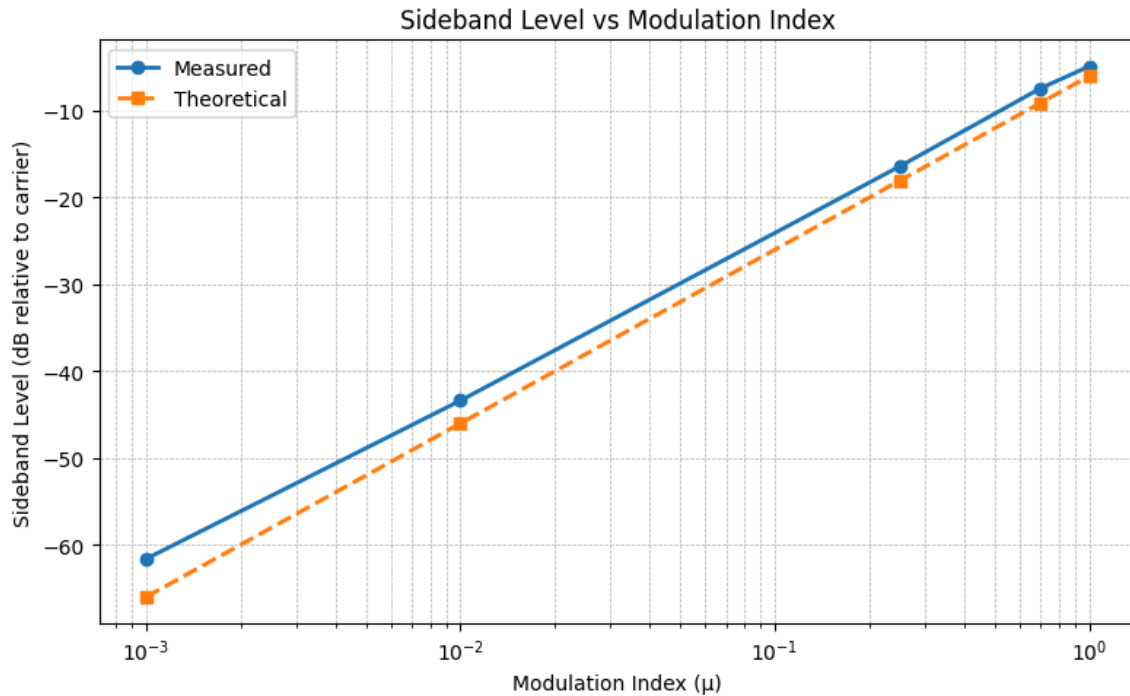
4.1.4 Using the values you determined in the previous question, plot the sideband level versus modulation index on a logarithmic plot for both the measured and the theoretical values

```
# Data
mu_values = np.array([0.001, 0.01, 0.25, 0.7, 1.0])
measured_levels = np.array([-61.63, -43.4, -16.41, -7.44, -4.9])
theoretical_levels = 20 * np.log10(mu_values / 2)

# Plot
plt.figure(figsize=(8, 5))
plt.semilogx(mu_values, measured_levels, 'o-', label='Measured', linewidth=2)
plt.semilogx(mu_values, theoretical_levels, 's--', label='Theoretical',
linewidth=2)

# Labels and grid
plt.xlabel("Modulation Index ( $\mu$ )")
plt.ylabel("Sideband Level (dB relative to carrier)")
plt.title("Sideband Level vs Modulation Index")
plt.grid(True, which='both', linestyle='--', linewidth=0.5)
plt.legend()
plt.tight_layout()
```

```
# Show plot  
plt.show()
```



4.1.5 Set the modulation index to $\mu = 0.7$ once more and change the modulating signal frequency to $f_m = 500$ Hz

To observe the AM signal in time domain, the spectrum analyser was configured in **zero span mode** following the procedure described in Section 2.2.1.5 of the lab manual:

1. The signal was tuned to the center frequency of the passband.
2. Both the resolution bandwidth (RBW) and video bandwidth (VBW) were set to **300 kHz**, which is more than twice the modulation frequency to avoid attenuation of sidebands.
3. The analyser was switched to **zero span mode**.
4. Linear display mode was selected, and the sweep time was adjusted to show several cycles of the demodulated waveform.

Configuration Settings:

- Trigger: *Video*
- Span: $476.1 \mu\text{V}$
- Sweep Time: 5 ms
- Measured Period: $T = 2$ ms

The measured period of the demodulated waveform was **2 ms**, which aligns with the expected theoretical period based on the modulation frequency:

$$T = \frac{1}{f_m} = \frac{1}{500} = 2 \text{ ms}$$

Sanity Check, if measured frequency matches out expectation

```
T_p = 2e-3 # [s]
f_meas = 1 / T_p

print(f"{f_meas:.1f} Hz")
```

500.0 Hz

Measurement Screenshots

The following screenshots show the configuration used to capture the AM signal in zero span mode:

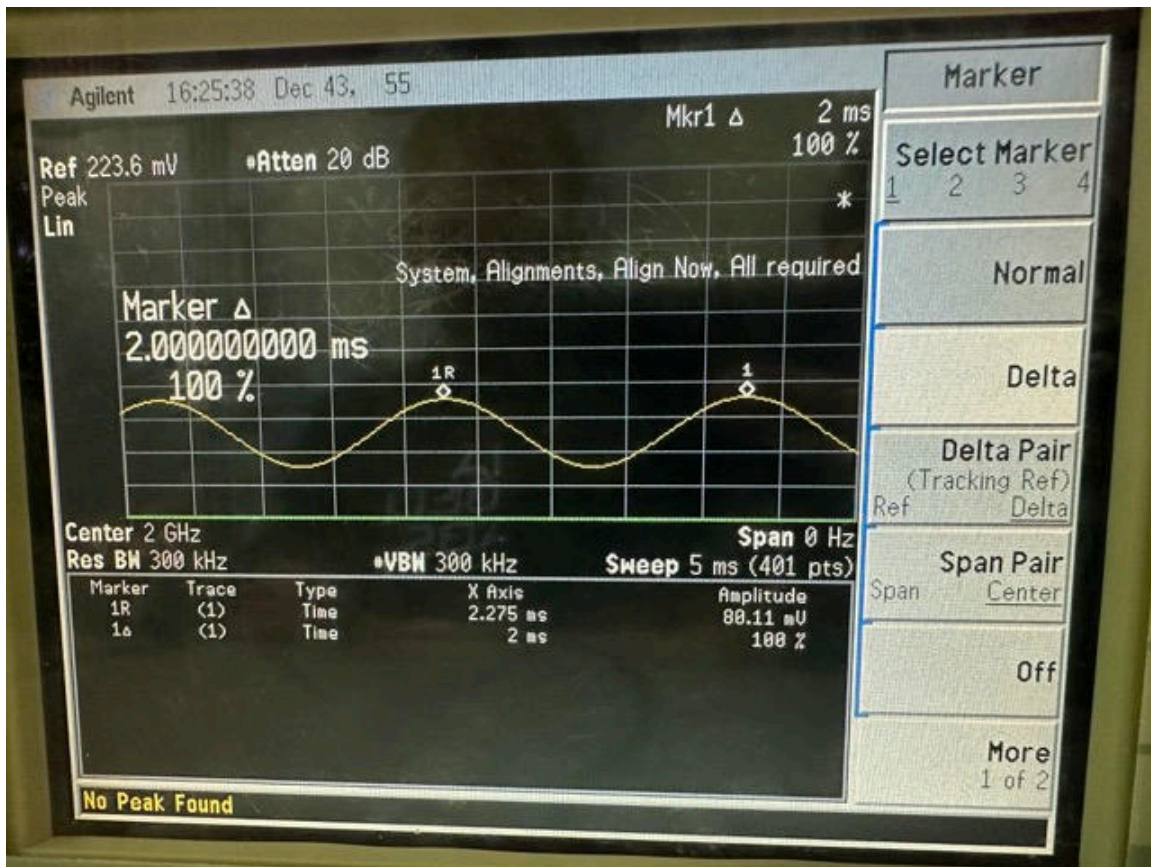


Figure 5: Spectrum analyzer in zero span mode showing a demodulated signal with a period of 2 ms.

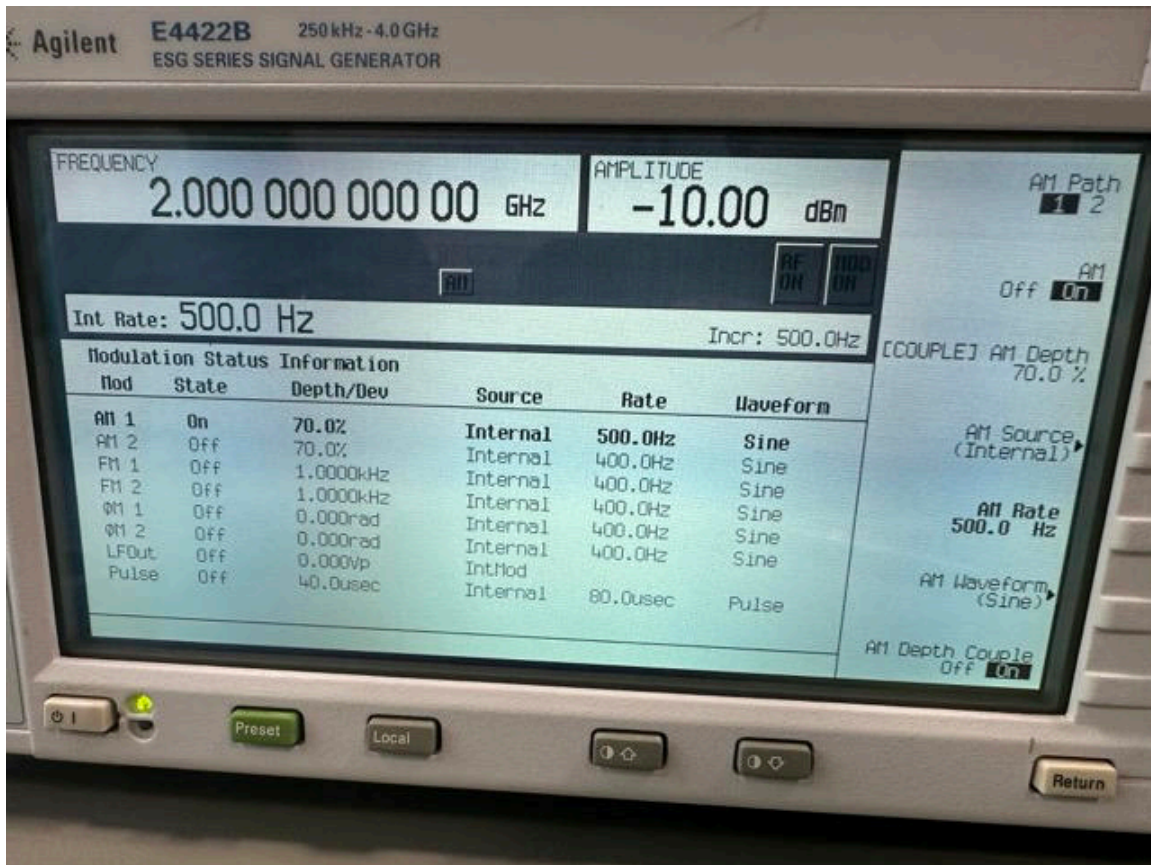


Figure 6: Signal generator settings with a 2 GHz carrier, -10 dBm output, 500 Hz modulation frequency, and 70% AM depth.

4.2 Task 2 Angle Modulation

4.2.2.1 & 4.2.2.2

Used Spectrum Analyzer Settings

- $RBW = 3kHz$
- Centre frequency $2GHz$
- Span $180kHz$
- Number of points: 400

4.2.2.3 Questions

Describe the spectrum you can see on the spectrum analyser.

We can observe a “comb” line spectrum with stronger symmetrical sidelobes. The carrier power is noticeably lower than the adjacent sidelobes. The sidelobes are equally spaced apart by $\approx 24.3kHz$

What type of angle modulation is the signal in Question 1?

Wideband FM modulation as $1 \leq \beta < 10$

What is the bandwidth of the signal displayed on the spectrum analyzer? How does this compare with the theoretical bandwidth value?

The spectrum observed on the analyzer showed that the sidebands were stronger than the carrier component. This is expected when the modulation index is near $\beta = 2.4$, which is close to a zero of the first-order Bessel function of the first kind, causing suppression of the carrier.

Given that $\beta = 2$, the signal falls within the range of wideband angle modulation (where $1 < \beta < 10$). According to Carson's Rule, the theoretical bandwidth is:

$$B_{FM} = 2B(\beta + 1) = 2 \cdot 25 \text{ kHz} \cdot (2 + 1) = 150 \text{ kHz}$$

The spectrum analyzer shows a value of $BW \approx 100.118 \text{ kHz}$, which is reasonably close, considering practical resolution and filter effects. The picture of the spectrum is available in Figure 7.

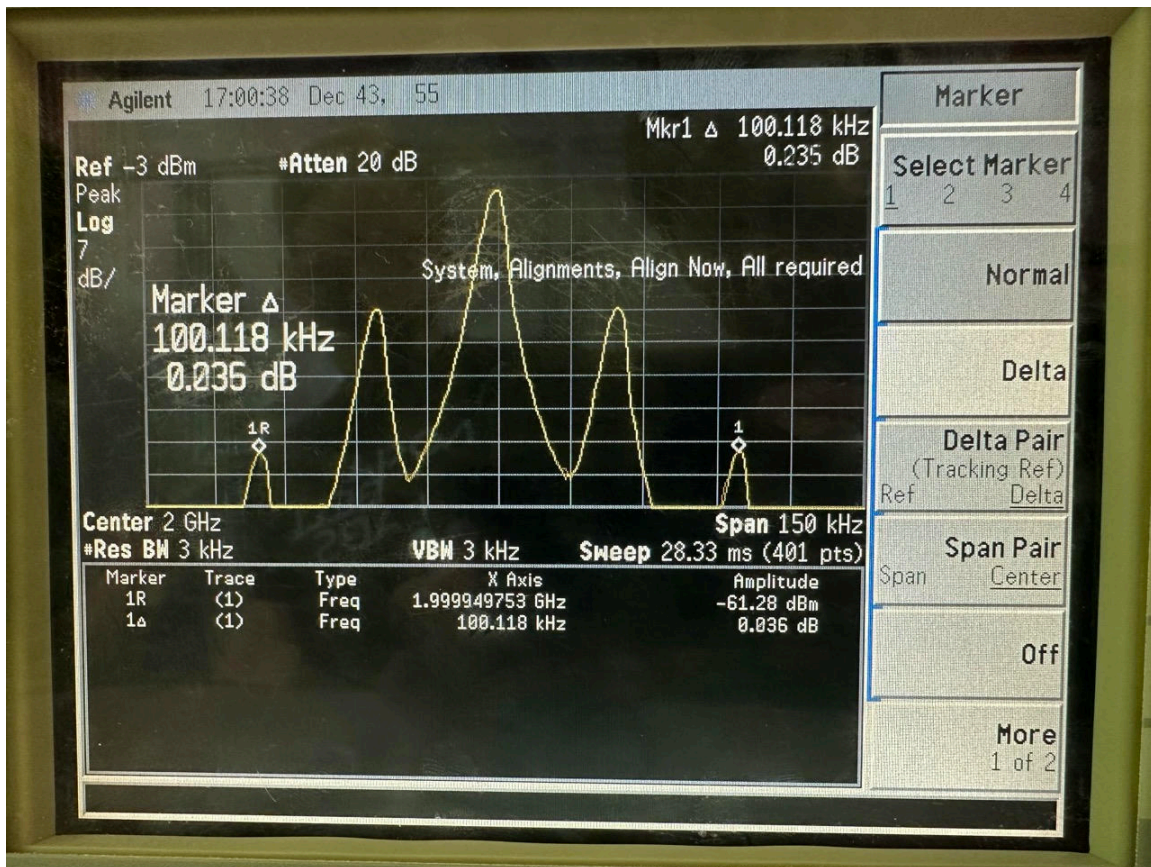


Figure 7: Spectrum analyzer output showing the FM-modulated signal for $\beta = 2$.

4.2.2.4 Change the modulation index to $\beta = 0.1$ and answer the following questions:

Describe the spectrum you can see on the spectrum analyser

The spectrum shows one high peak at the carrier and very close to it the two adjacent peaks. The distance between the main and first sidelobe is $\approx 24.75 \text{ kHz}$. The next sidelobes are considerably smaller.

What type of angle modulation is the signal in Question 4?

This is a narrowband FM signal, as $\beta < 1$

What is the bandwidth of the signal displayed on the spectrum analyser? How does this compare with the theoretical bandwidth value?

The observed bandwidth, observed in Figure 8 was approximately 50 kHz ($\approx 2 \cdot 24.75 \text{ kHz}$), covering the carrier and one sideband on each side. The theoretical bandwidth using Carson's Rule for narrowband FM is:

$$B_{\text{FM}} \approx 2f_m = 2 \cdot 25,000 = 50 \text{ kHz}$$

The measured and theoretical bandwidth values match well, confirming the validity of the approximation for small values of β .

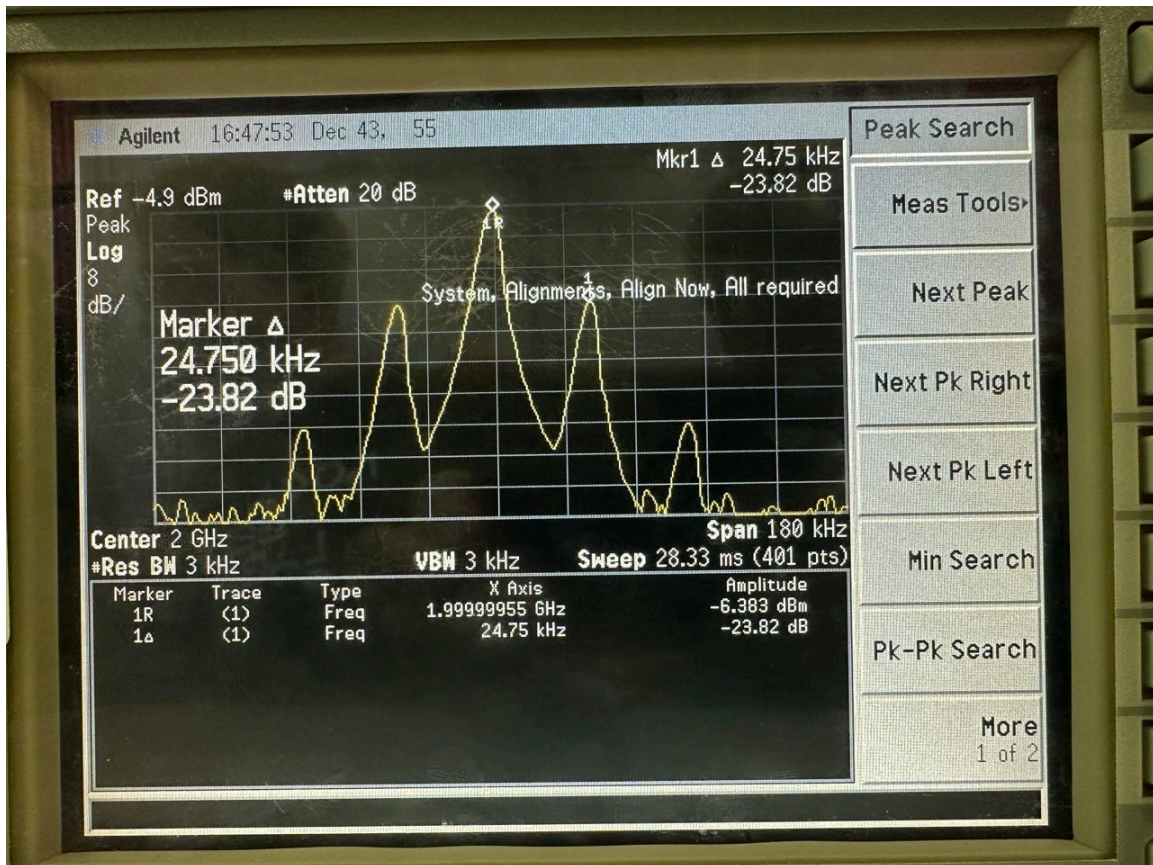


Figure 8: Spectrum analyzer output showing the FM-modulated signal for $\beta = 0.1$.

4.2.2.5 Change the modulation index to β , in order to answer the following questions

At what value of β does the carrier component no longer appear in the spectrum?

The carrier would disappear according to the Besselfunction (as seen in Figure 9) at $J_0(2.4) = 0$.

We could very closely match this with a value of $\beta = 2.42$

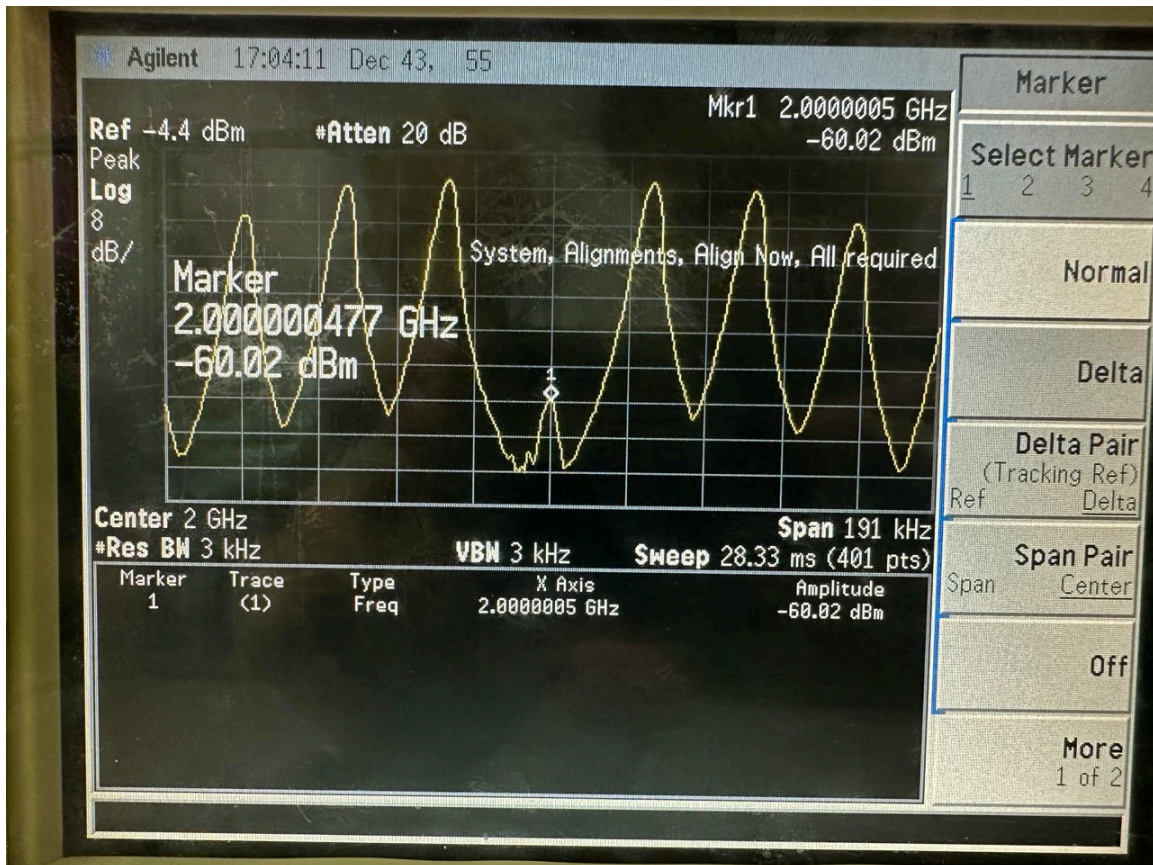


Figure 9: Spectrum analyzer output showing the FM-modulated signal for $\beta = 2.42$. Notice that the carrier is not visible.

At what value of β does the 1st sideband components no longer appear in the spectrum?

The Besselfunction would predict this to happen at $J_1(4) = 0$. This was possible to recreate with a value of $\beta = 3.86$

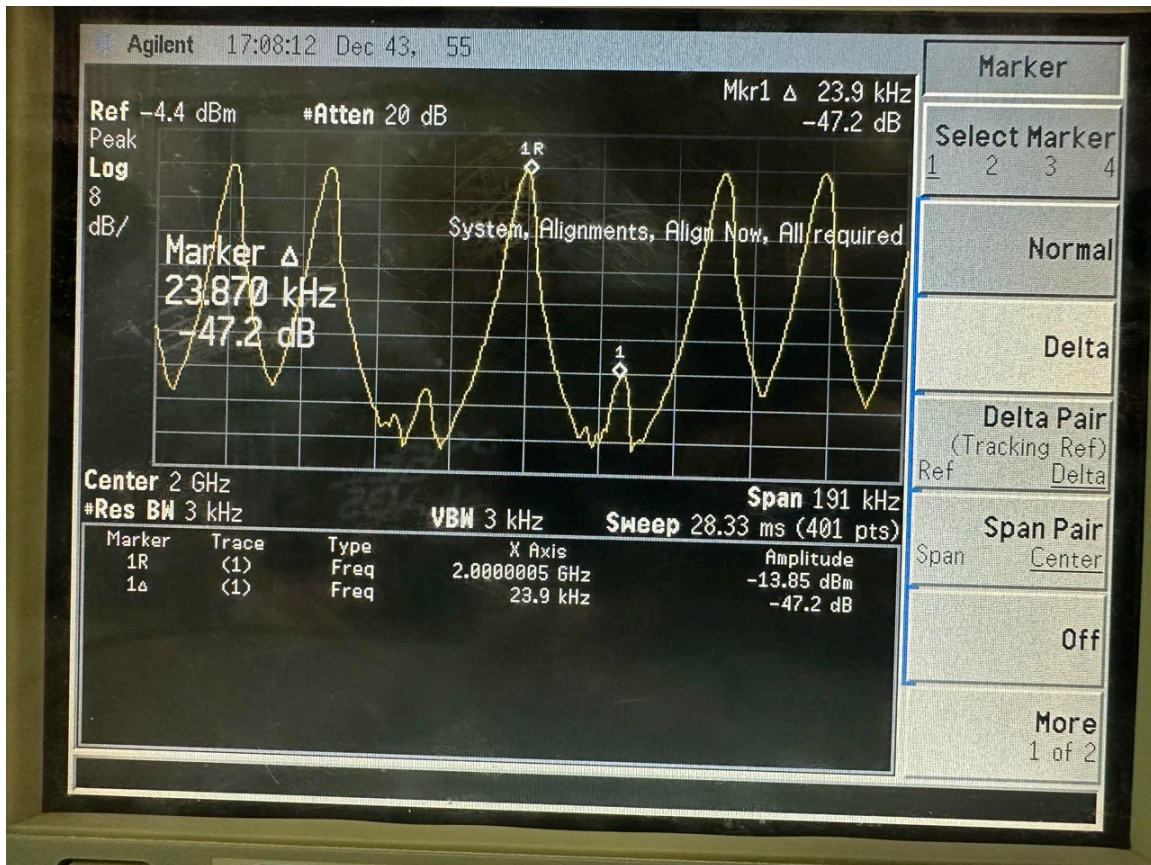


Figure 10: Spectrum analyzer output showing the FM-modulated signal for $\beta = 3.86$. Notice that the first sideband component is not visible.

What is the next value of β where the carrier component again no longer appears in the spectrum?

The Besselfunction would predict this to happen at $J_0(\approx 5.6) = 0$. With a value of $\beta = 5.33$ we could accomplish a minimum carrier power. As seen in Figure 11 showing the spectra and Figure 12 showing the settings of the signal generator.

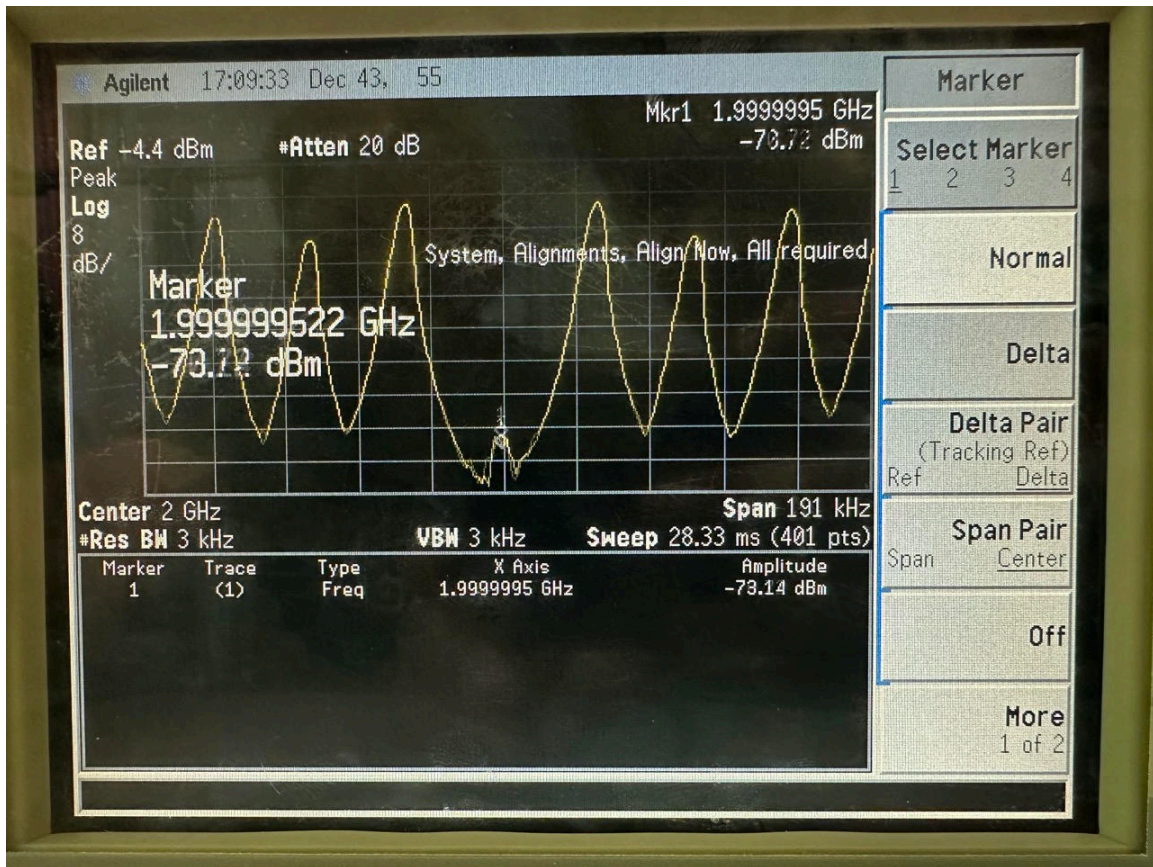


Figure 11: Spectrum analyzer output showing the FM-modulated signal for $\beta = 5.33$. Notice that the carrier is not visible.

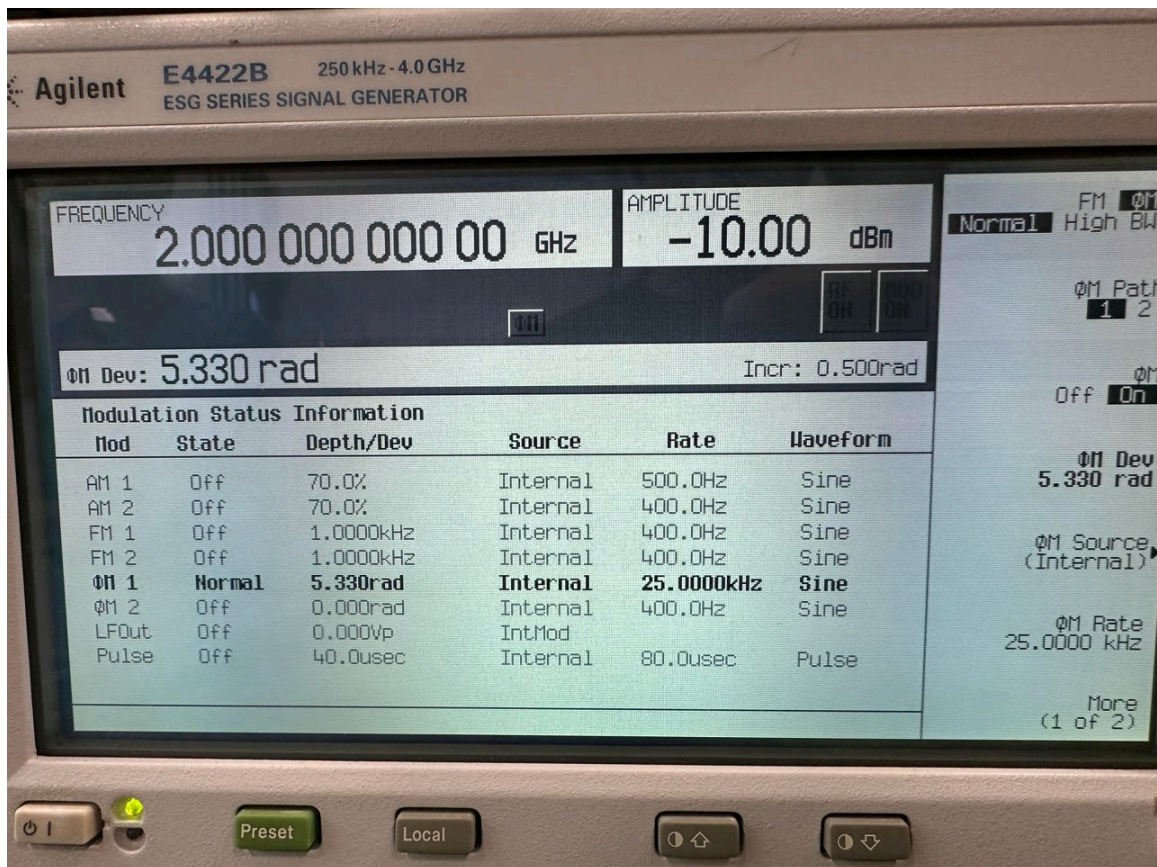


Figure 12: Settings from the signal generator for the next value of β where the carrier is not visible.

4.2.2.6 Change the modulation index to $\beta = 12$ and the modulating frequency to $f_m = 1$ kHz

The parameters in the signal generator were adjusted, yielding the spectra from Figure 13.

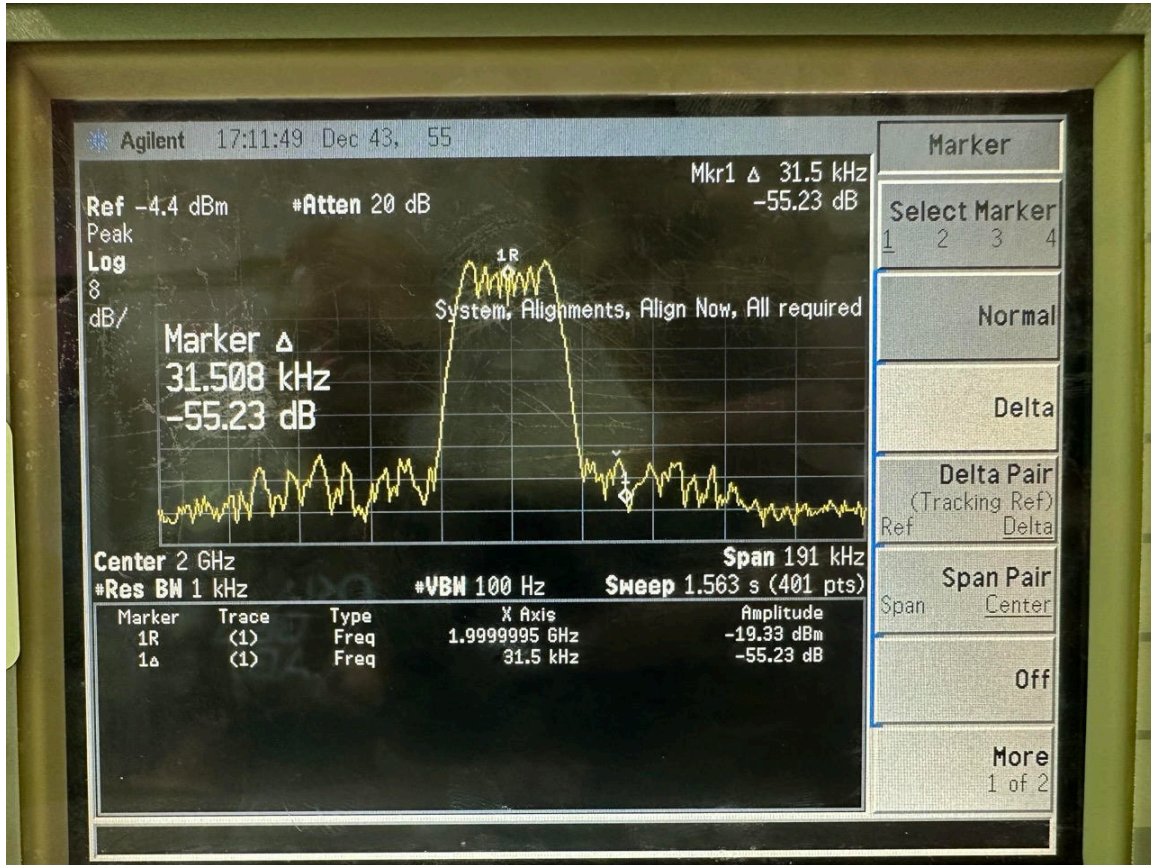


Figure 13: Spectrum analyzer output showing the FM-modulated signal for $\beta = 12$.

Describe the spectrum you can see on the spectrum analyser

We can see one big symmetrical lobe around the carrier. This big lobe has a small “comb” on top of it. There are no other sidelobes.

What type of angle modulation is the signal in Question 6?

We can see a ultra-wideband FM signal as $\beta > 10$

What is the bandwidth of the signal displayed on the spectrum analyser? How does this compare with the theoretical bandwidth value?

As it can be seen in Figure 14, the bandwidth is 25 25.8 kHz. Theoretical FM bandwidth can be estimated using Carson’s Rule:

$$B_{FM} = 2f_m(\beta + 1) = 26 \text{ kHz}$$

The measured and theoretical values are therefore consistent within practical accuracy.

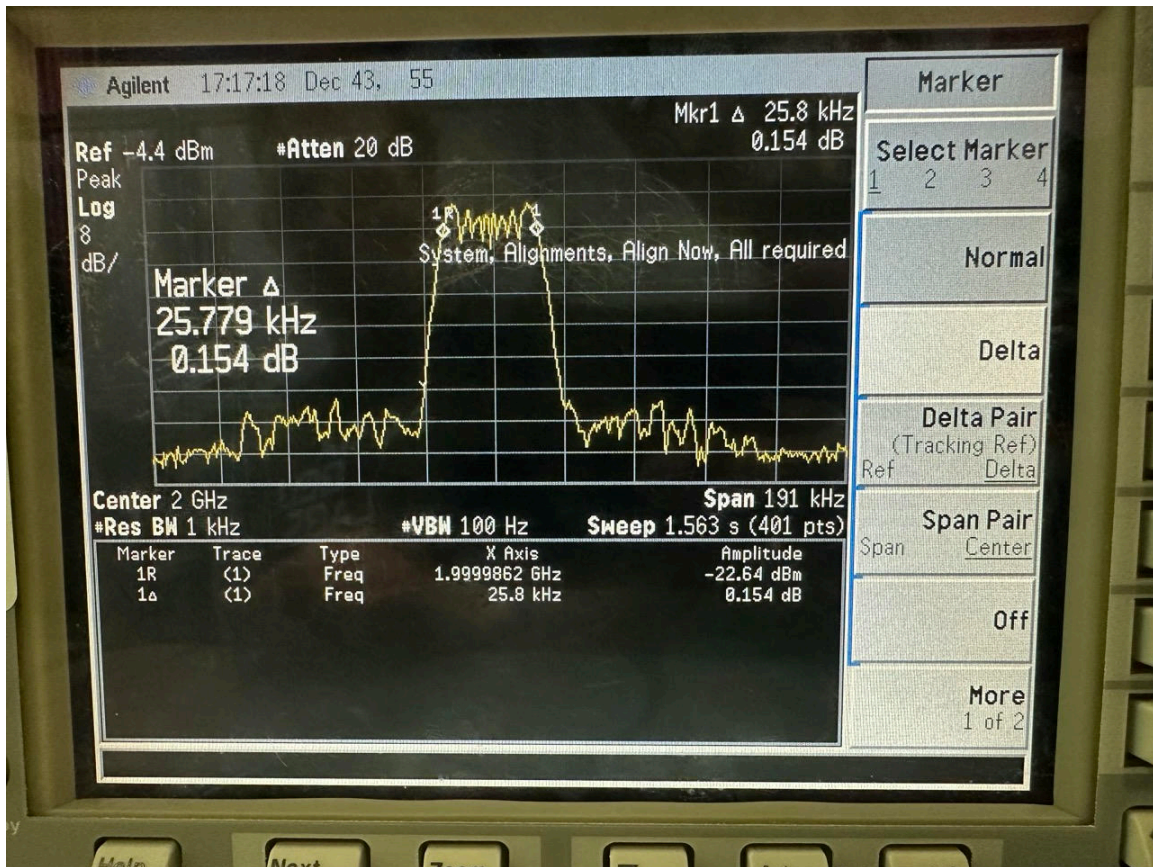


Figure 14: Bandwidth of the FM-modulated signal for $\beta = 12$.

4.3 Task 3: Pulse spectrum

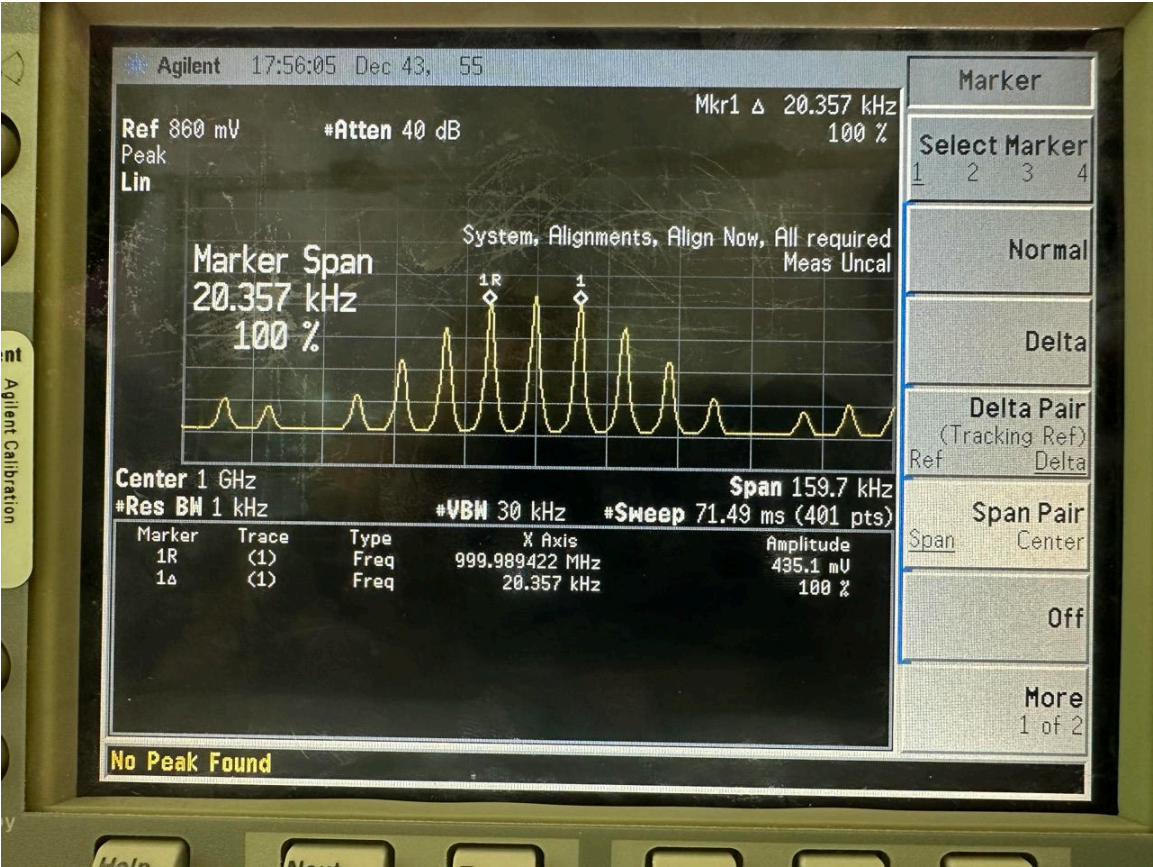


Figure 15: Envelope spectrum. Is is worth noting that the distance between lines is $25.357\text{ kHz} / 2 \approx 10\text{ kHz}$ as observed by the delta marker.

- Carrier frequency $f_c = 1\text{GHz}$
- Output power $P_{out} = 10\text{dBm}$
- Pulse period $T = 100\mu s$
- Duty Cycle 20%

4.3.3.2

\text{Insert answers}}

Spectrum Analysis Settings

The following settings were used to capture the envelope and line spectra of a pulse-modulated signal. These values were obtained from Figures 15, 16, and 17.

Spectrum Analysis Settings

Setting	Envelope Spectrum	Line Spectrum
RBW (Resolution Bandwidth)	1 MHz	1 kHz

Setting	Envelope Spectrum	Line Spectrum
Video Bandwidth	30 kHz	30 kHz
Centre Frequency	1 GHz	1 GHz
Span	159.7 kHz	97 kHz
Number of Points	401	401

Envelope Spectrum

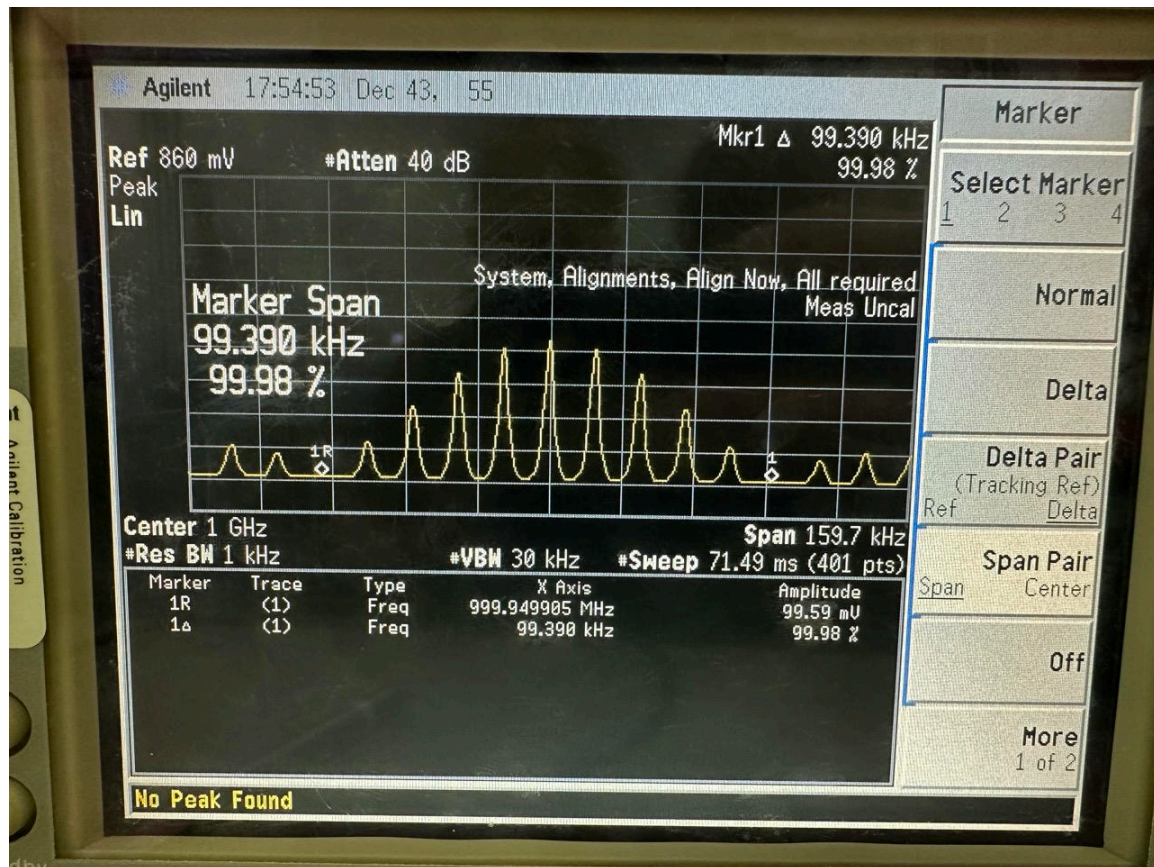


Figure 16: Envelope spectrum. It is worth noting that the distance of the main lobe is $\approx 100\text{kHz}$ hence the zero spacing is approximately 50 kHz.

Line Spectrum

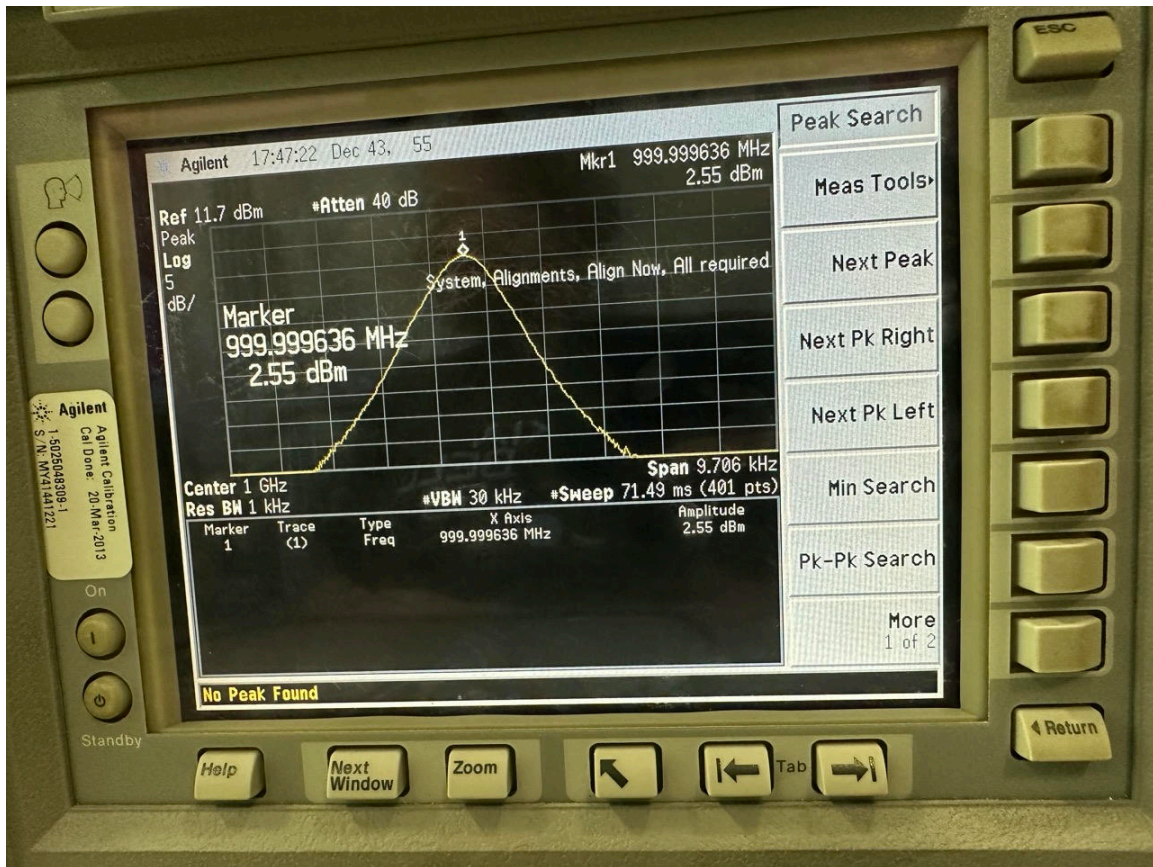


Figure 17: Pulse spectrum. It is worth noting that the distance of the main lobe is $\approx 100\text{kHz}$ hence the zero spacing is approximately 50 kHz.

Zero Span Mode

The spectrum of a rectangular pulse follows a sinc envelope. Evaluating the sinc function at this frequency:

$$\text{sinc}(\pi f\tau) = \text{sinc}(\pi \cdot 10,000 \cdot 20 \times 10^{-6}) = \text{sinc}(0.628) \approx 0.21$$

Converting this to decibels:

$$20 \log_{10}(0.21) \approx -13.5 \text{ dB}$$

This theoretical result closely matches the measured value, confirming that the spectral envelope follows the expected sinc-shaped profile. The difference is attributed to non ideal conditions during the measurements, like the losses and possible mismatching from the cables as well as noise.



Figure 1: 4-3-4-sinc-sa.jpeg

Figure 19: Sinc spectrum. From the marker the amplitude difference between the main lobe and the first lobe is obtained via the marker.

4.3.3.5 Now study the influence of an increasing RBW on the amplitude for both types of spectra.

For the line spectrum: What happens to the amplitude of the line spectrum when the RBW is increased?

$\text{\textcolor{red}{\text{\text{Insert answers}}}}$

For the Envelope Spectrum

What happens to the amplitude of the envelope spectrum when the RBW is increased?

$\text{\textcolor{red}{\text{\text{Insert answers}}}}$

Explain this behaviour. Support your explanation with a calculation

$\text{\textcolor{red}{\text{\text{Insert answers}}}}$

Further increase the RBW – at what point does the behaviour/trend you described in the previous question no longer happen? Why?

$\text{\textcolor{red}{\text{\text{Insert answers}}}}$

4.4 Task 4: Mixer

- Local Oscillator (LO) frequency: $f_{LO} = 4$, $\text{\textcolor{red}{\text{\text{Insert answers}}}}$
- LO power: $P_{LO} = 7$, $\text{\textcolor{red}{\text{\text{Insert answers}}}}$
- Radio Frequency (RF) frequency: $f_{RF} = 2.9$, $\text{\textcolor{red}{\text{\text{Insert answers}}}}$
- RF power: $P_{RF} = 0$, $\text{\textcolor{red}{\text{\text{Insert answers}}}}$

4.4.2

- Start Frequency $> 0\text{Hz}$ as a DC-Block was used
- Stop Frequency $\approx 10.5\text{GHz}$
- RBW 1MHz
- Number of points 401

4.3.3 Write down the frequencies of all the harmonics and intermodulation products and compare the resultant frequencies with the theoretically predicted harmonics.

With the help of our tool from the preliminary, these frequencies can be expected:

Mixer Analysis: LO = 4.0 GHz, RF = 2.9 GHz

=====

Fundamental:

2.90 GHz	f_{RF}	Order 1
4.00 GHz	f_{LO}	Order 1

Harmonics:

5.80 GHz	$2f_{RF}$	Order 2
8.00 GHz	$2f_{LO}$	Order 2
8.70 GHz	$3f_{RF}$	Order 3
12.00 GHz	$3f_{LO}$	Order 3

Sum Diff:

1.10 GHz		$f_{\{LO\}} - f_{\{RF\}}$		Order 2
6.90 GHz		$f_{\{LO\}} + f_{\{RF\}}$		Order 2

Intermodulation:

1.80 GHz		$2f_{\{RF\}} - f_{\{LO\}}$		Order 3
5.10 GHz		$2f_{\{LO\}} - f_{\{RF\}}$		Order 3
9.80 GHz		$f_{\{LO\}} + 2f_{\{RF\}}$		Order 3
10.90 GHz		$2f_{\{LO\}} + f_{\{RF\}}$		Order 3

Compared with the screenshot from the last exercise, we can see: - Fundamentals are (2) and (3) are where we expect them - Second order harmonics $f_{LO} \pm f_{RF}$ are (1) at $\approx 1.1GHz$ and (4) at $\approx 6.9GHz$ - Second order harmonics at $2f_{LO} \approx 8GHz$ at (7) - Second order harmonics at $2f_{RF} \approx 5.8GHz$ could be one of the two peaks between (8) $4.7GHz$ and (4) $6.9GHz$ - Third order harmonics at $f_{LO} + 2f_{RF}$ is visible (6) at $\approx 9.8GHz$ - Third order harmonics at $2f_{LO} + f_{RF}$ was outside of the measurement range - Third order harmonics at $2f_{LO} - f_{RF}$ could be the peak next to (8) or even be (8) $\approx 4.7GHz$, but this would be stronger than expected deviation

So overall our expectations for the location in the spectrum are fulfilled.

4.3.4 Answer the following questions and measure the isolation between the following ports

We are referncing this datasheet

a) How should the third port be terminated when measuring the isolation between the two remaining ports?

It should be terminated as a matched load therefore with an impedance $Z_0 = 50\Omega$, because further reflections caused by mismatched load would skew the result.

b) What is the conversion loss (dB)? Compare it with the value on the datasheet

Although the datasheet doesnt have values for the exact same frequencies, but we can assume, because all values at that frequency-range are around $8.41dB$ and $9.22dB$. $\text{\textcolor{red}{where}}$

c) What is the LO-IF isolation (dB)? Compare it with the value on the datasheet.

$\text{\textcolor{red}{\text{Insert answers}}}$

d) What is the LO-RF isolation (dB)? Compare it with the value on the datasheet.

$\text{\textcolor{red}{\text{Insert answers}}}$